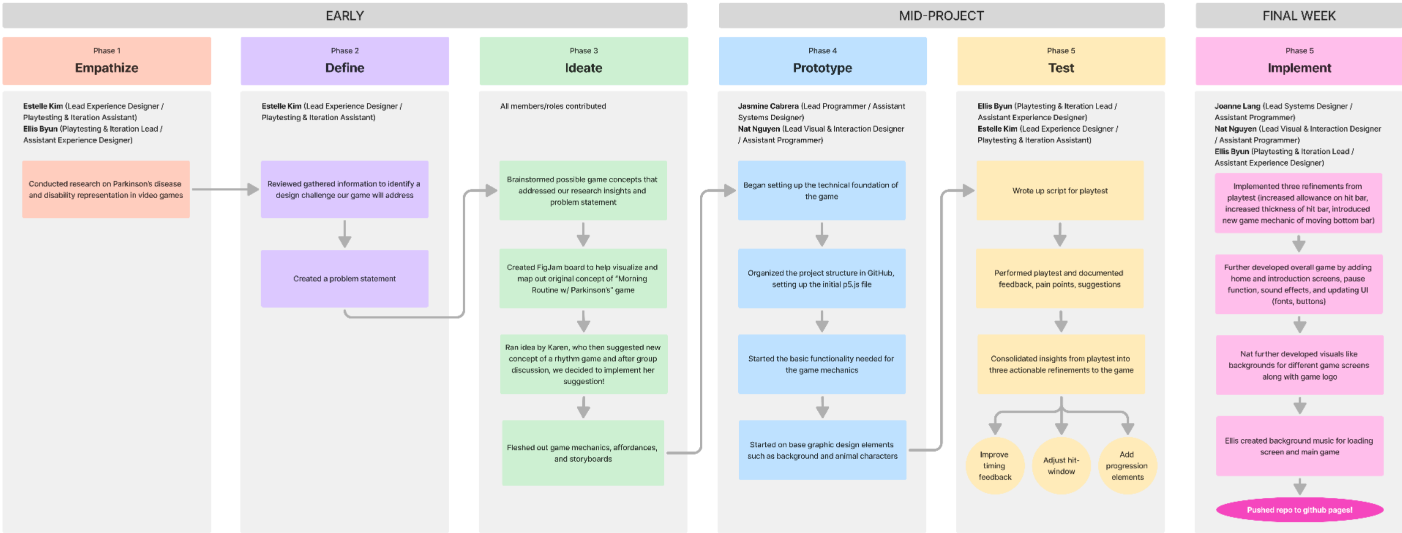
Process & Decision Documentation

Process Overview Visualization (Group Work Only)

https://www.figma.com/board/m9n2WSNvKWZ882ZpkUhfc1/Untitled?node-id=0-1&t=LKb7zbT8bUtQtfLb-1



Project/Assignment Decisions

- Estelle led the adaptation of our new game concept suggestion from Dr. Cochrane via Teams
    - What changed: Our initial game concept was a "Morning Routine with Parkinson's game, but after discussing the idea with Dr. Cochrane, we changed our concept to a rhythm game
    - Why: Dr. Cochrane pointed out various areas in which our original game concept fell short, the biggest problem being that our game felt too much like a simulation and lacked real game mechanics. After team discussion, we ultimately felt it would be smart to move forward with the rhythm game to have clear game mechanics that tied in with Parkinson's.
- Nat: Visual designs were inspired by other synth-style rhythm games, like Bemuse.ninja and MuseDash, with added neon styles and kitty/bunny pixel art graphics
    - Initially created a plain galaxy theme but switched to more of cyberpunk neon theme (adding buildings, neon accents)

    - Considered full-body character designs, but opted for simple pixel bunny and kitty to reduce distraction and keep focus on gameplay
    - Chose pixel art for most visuals to emphasize retro/rhythm-game feel
    - Bright neon, pixelated logo with musical accents to match UI
- Ellis: Iteration planning based on playtesting feedback and produced background music inspired by tech-y, digital sounds typically seen in rhythm games, adding to the cyberpunk theme
    - Took notes on player experience and feedback during playtesting; synthesized observations into 3 main improvements for the next mid-term iteration (improve timing feedback, adjust hit window and error feedback, add progression elements to increase engagement)
    - Made idle BGM softer, slower, and more relaxed
    - Gameplay BGM is faster, adding to the intensity of the game
- Joanne and Jasmine decided to use the Google Drive app to collaboratively code
    - The group considered adding an Instability Bar that would fill up as challenges came across the screen and the player tried to adapt to them while playing the rhythm game, but scrapped this due to unnecessary complexity
    - Jasmine coded the base architecture including the main game loop, player input handling, and core object interactions.
    - Building stage 1 functionality to make the prototype playable for initial testing, ensuring all basic mechanics (movement, collision detection, scoring, and level progression) were functional. Developed modular JavaScript structure to support future feature additions, making it easier for the team to implement mechanics and assets in later iterations. Ensured the prototype was playtest-ready, enabling peers and instructors to interact with the game and provide meaningful feedback for iteration.
    - The group considered adding a BrainFreeze mode where blocks were “frozen” and had to be hit 3 times with key inputs in order to be registered as a successful hit; this was scrapped after feedback from Midterm Showcase to keep things simple and focus on the rhythm mechanic
    - Joanne updated the code based on feedback from playtesting (bigger hit bar for greater hit allowance, making level progressively harder) and UI elements
- Joanne published to GitHub and launched to GitHub Pages

Role-Based Process Evidence

Entry #1: Disability Framing + Research and Game Concept Ideation

Name: Estelle Kim

Role(s): Lead Experience Designer, Playtesting & Iteration Assistant

Primary responsibility for this work: Responsible for disability framing + research on disability representation and creating empathy in games

Goal of Work Session

- Conduct various forms of secondary research (online articles, peer-reviewed articles, videos, lecture slides)
- Gather information both on Parkinson's disease and disability representation in games
- Synthesize research findings to create problem statement and help with game concept ideation
- Ensure information was thoroughly referenced with proper citations
- Initiate and lead game concept ideation sessions with team

Tools, Resources, or Inputs Used

- ChatGPT 5.2
- Lecture Notes (Week 4 - part 1)
- External resources (online articles, videos, conference paper, etc.) - found in *references*

GenAI Documentation

If GenAI was used (keep each response as brief as possible):

Date Used: Feb 27, 2026

Tool Disclosure: ChatGPT 5.2

Purpose of Use: To provide a starting point for research, understanding exactly what information is crucial to gather and include and how to organize my notes.

Summary of Interaction: The AI produced generated possible sections of research I should focus on, listing exactly what topics and points of information should be included. It also suggested how I should format my notes so that they are organized, and easy to navigate.

Human Decision Point(s): Did not follow the suggested information organization exactly–some sections were reordered based off what made more sense in my mind, "Teaching Outcome Clarity" suggested section was not included as I found it irrelevant to our specific research, and I added additional sections that were not suggested by AI as it did not consider important themes/topics that were discussed in class, such as "Disability Tourism".

Integrity & Verification Note: Before following the AI's suggestions on how to collect and start my research, I ensured to cross-reference the assignment outline to ensure that the information it recommended I gather was relevant to the assignment and our team's goal.

Scope of GenAI Use: Actual research and data collection was all done without the use of AI; game concept ideation was all done without AI, collaboratively with the team.

Limitations or Misfires: When suggesting what information to collect for research, the AI was not cognizant of important topics and themes that were specifically talked about in class.

Entry #2: Technical Programmer and Early Implementation

Name: Jasmine Cabrera

Role(s): Lead Programmer (Technical Implementation), Assistant Systems Designer

Primary responsibility for this work: Responsible for stage one of coding which includes setting up the code architecture and basic functionality of the game. I also supported the system designer by helping translate gameplay ideas and mechanics into technical logic that could be implemented in code.

Goal of Work Session

The goal during this phase of assignment was to begin setting up the technical foundation of the game. This included organizing the project structure in GitHub, setting up the initial p5.js file, and starting the basic functionality needed for the game mechanics. Another goal was to work with the systems designer to understand how the gameplay mechanics would function so that I could build the necessary code structure that would support those mechanics later in development.

Tools, Resources, or Inputs Used

- GitHub
- p5.js coding environment

- ChatGPT (for brainstorming technical structure and trouble shooting small coding questions)
- Prior drafts or code
- Teams (discussion with system designer and visual designer)
- Lecture notes on coding structure and game mechanics

GenAI Documentation

If GenAI was used (keep each response as brief as possible):

Date Used: Feb 26, 2026

Tool Disclosure: ChatGPT 5.2

Purpose of Use: Used mainly for brainstorming, clarifying code mechanics and how the game could be split into different technical stages so multiple team members could work on the project.

Summary of Interaction: The AI helped explain how the coding process for a game could be divided into stages such as code architecture, gameplay mechanics, and interface systems.

Human Decision Point(s): Our group adapted the suggested stages but adjusted them to our project needs. We decided that my role would focus on stage 1 (code architecture) while other members would focus on game mechanics and UI systems later.

Integrity & Verification Note: The suggestions from the AI were discussed with the group and compared with what we learned in class about programming workflow and game design. We only used the AI suggestions as guidance and modified them to fit our project.

Scope of GenAI Use: The AI did not write our comments, it was used to help our code, design mechanics, and brainstorm how responsibilities can be divided for group work.

Limitations or Misfires: Some of the suggestions were too general and did not fully match the tools we were using (p5.js and GitHub). Because of this, we had to adjust the suggestions and simplify them to better fit our project workflow.

Summary of Process (Human + Tool)

Our group discussed dividing up the roles first since multiple people would be contributing to the game. Because GitHub makes it difficult for multiple people to work on the same files, we wanted to organize the development. Being lead programmer I was just responsible for early stage of coding, I started by setting up the project structure and organizing the basic files needed for the game. I focused on creating a foundation that would allow the systems designer and visual designer to later add gameplay mechanics and interface elements. Throughout this stage, I communicated with the system and visual designers to understand how the core gameplay mechanics would function. Which helped me ensure that the layout I was building would support the mechanics later in development.

Decision Points & Trade-offs

- Simplifying early code architecture

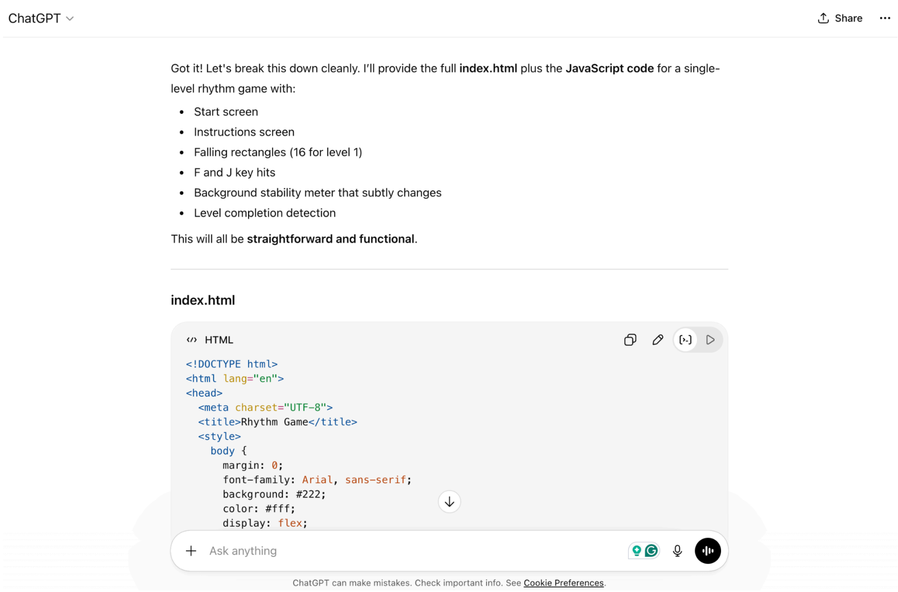
Initially, we considered creating a more complex system structure, but we decided to keep the architecture simpler so that other team members could easily understand and modify the code later. This made collaboration easier.

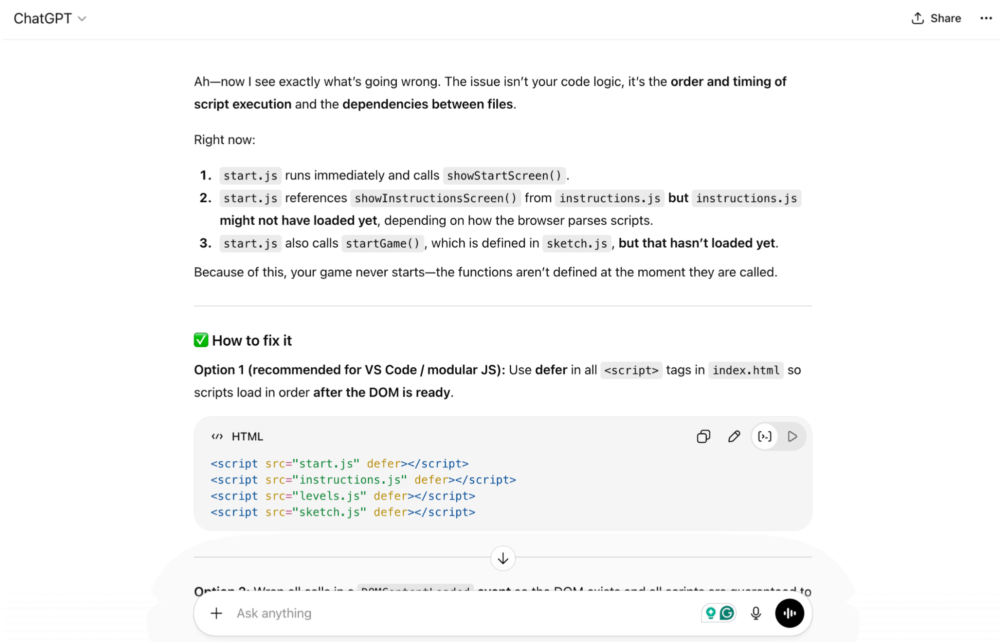
Verification & Judgement

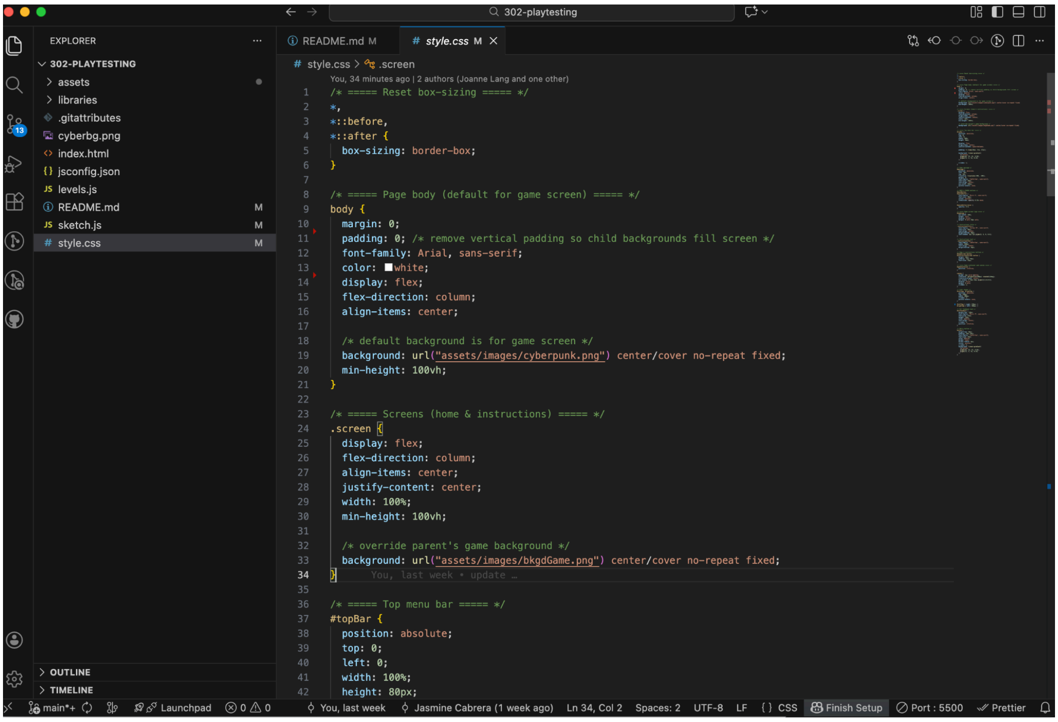
- Lots of playtesting
- Group in person/Online discussion
- Peer or instructor feedback
- Comparison with course concepts
- Re-reading assignment criteria

Limitations, Dead Ends, or Open Questions

One limitation during this stage was that some gameplay mechanics were still being finalized, which made it difficult to fully plan the code structure. Because of this, the architecture had to remain somewhat flexible so it could be adjusted later.







Entry #3: Playtesting Prep, Research Integration, and Audio Design

Name: Ellis Byun

Role(s): Playtesting & Iteration Lead, Assistant Experience Designer

Primary responsibility for this work: Responsible for preparing playtesting structure and documentation, supporting player experience design and disability framing research, and designing the audio tracks used in the game.

Goal of Work Session

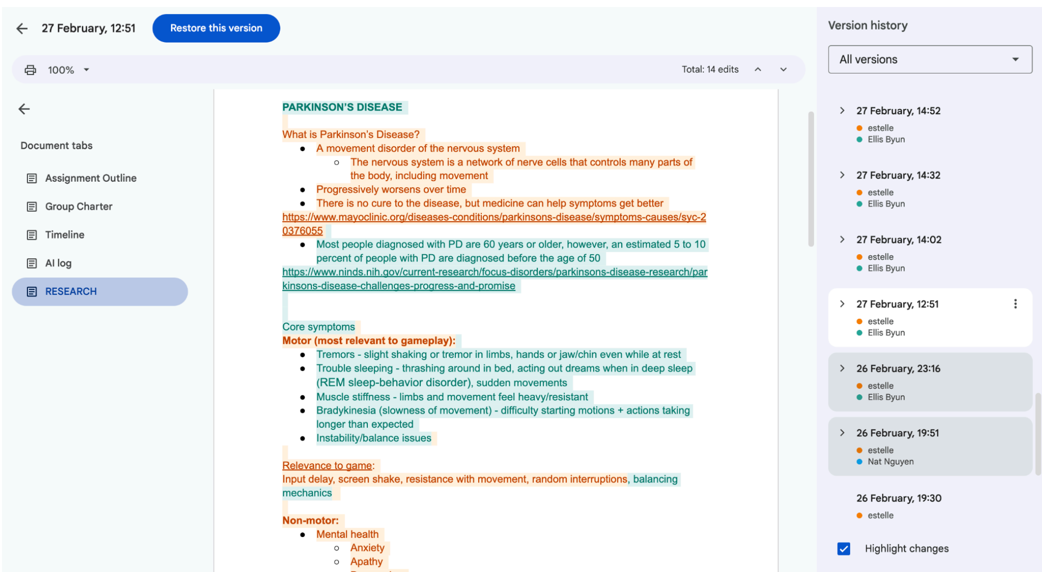
- Conducted secondary research (online articles, peer-reviewed sources, lecture materials) related to Parkinson's disease and movement disorders
- Reviewed course material on disability framing to ensure game concept avoids stereotypes or pity-based narratives
- Synthesized research findings into insights that could inform gameplay mechanics and player experience
- Contributed to team brainstorming sessions for potential game concepts based on research findings
- Helped translate research insights into mechanics that could be represented through rhythm-based game system
- Created audio tracks for the game using Soundtrap to support rhythm mechanics and pacing of levels
- Assisted in preparing playtesting documentation and identifying aspects of the prototype that should be evaluated during testing

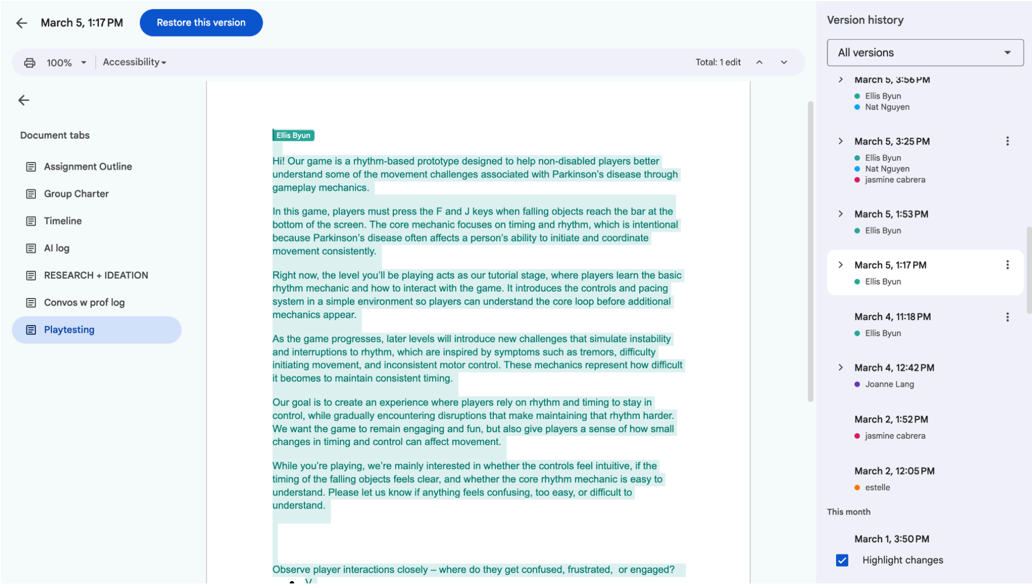
Tools, Resources, or Inputs Used

- Lecture Notes (Week 4 - part 1)
- Academic articles and online sources about Parkinson's disease
- Soundtrap (audio production)
- Team discussion and ideation sessions
- Playtesting feedback

GenAI Documentation

No AI was used.





March 5, 1:17 PM
Restore this version
100%
Accessibility
Total: 1 edit

Document tabs
Assignment Outline
Group Charter
Timeline
AI log
RESEARCH + IDEATION
Convos w prof log
Playtesting

Ellis Byun

Hi! Our game is a rhythm-based prototype designed to help non-disabled players better understand some of the movement challenges associated with Parkinson's disease through gameplay mechanics.

In this game, players must press the F and J keys when falling objects reach the bar at the bottom of the screen. The core mechanic focuses on timing and rhythm, which is intentional because Parkinson's disease often affects a person's ability to initiate and coordinate movement consistently.

Right now, the level you'll be playing acts as our tutorial stage, where players learn the basic rhythm mechanic and how to interact with the game. It introduces the controls and pacing system in a simple environment so players can understand the core loop before additional mechanics appear.

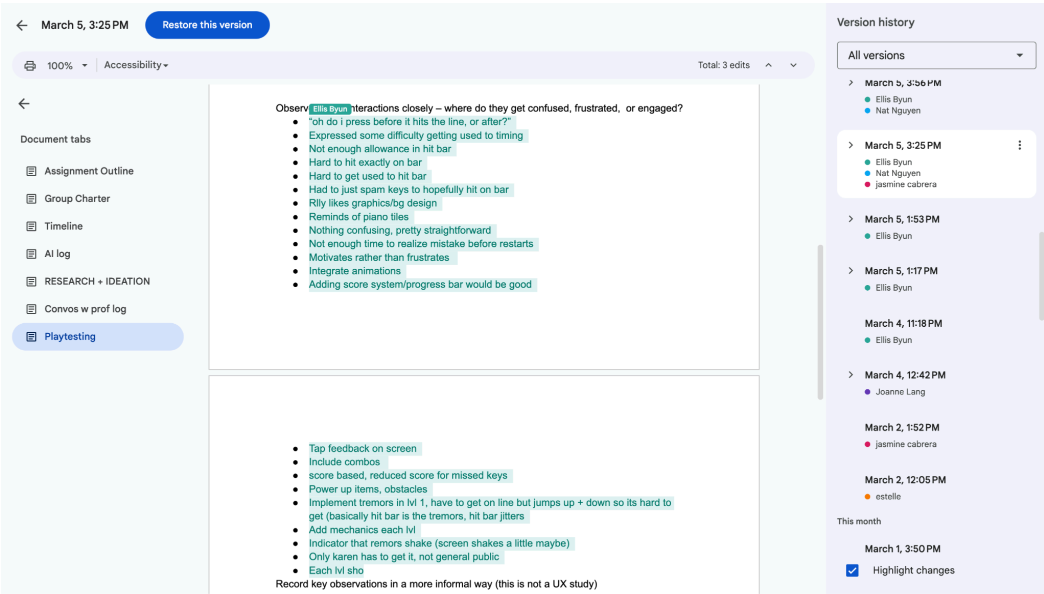
As the game progresses, later levels will introduce new challenges that simulate instability and interruptions to rhythm, which are inspired by symptoms such as tremors, difficulty initiating movement, and inconsistent motor control. These mechanics represent how difficult it becomes to maintain consistent timing.

Our goal is to create an experience where players rely on rhythm and timing to stay in control, while gradually encountering disruptions that make maintaining that rhythm harder. We want the game to remain engaging and fun, but also give players a sense of how small changes in timing and control can affect movement.

While you're playing, we're mainly interested in whether the controls feel intuitive, if the timing of the falling objects feels clear, and whether the core rhythm mechanic is easy to understand. Please let us know if anything feels confusing, too easy, or difficult to understand.

Observe player interactions closely – where do they get confused, frustrated, or engaged?

Version history
All versions
March 5, 3:56 PM
Ellis Byun
Nat Nguyen
March 5, 3:25 PM
Ellis Byun
Nat Nguyen
jasmine cabrera
March 5, 1:53 PM
Ellis Byun
March 5, 1:17 PM
Ellis Byun
March 4, 11:18 PM
Ellis Byun
March 4, 12:42 PM
Joanne Lang
March 2, 1:52 PM
jasmine cabrera
March 2, 12:05 PM
estelle
This month
March 1, 3:50 PM
Highlight changes



March 5, 3:25 PM
Restore this version
100%
Accessibility
Total: 3 edits

Document tabs
Assignment Outline
Group Charter
Timeline
AI log
RESEARCH + IDEATION
Convos w prof log
Playtesting

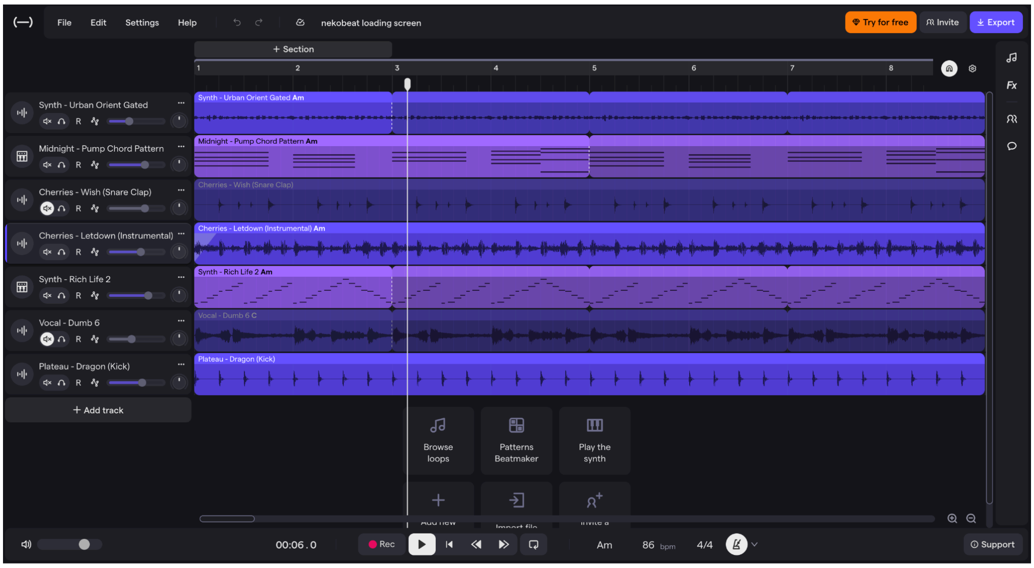
Observe player interactions closely – where do they get confused, frustrated, or engaged?

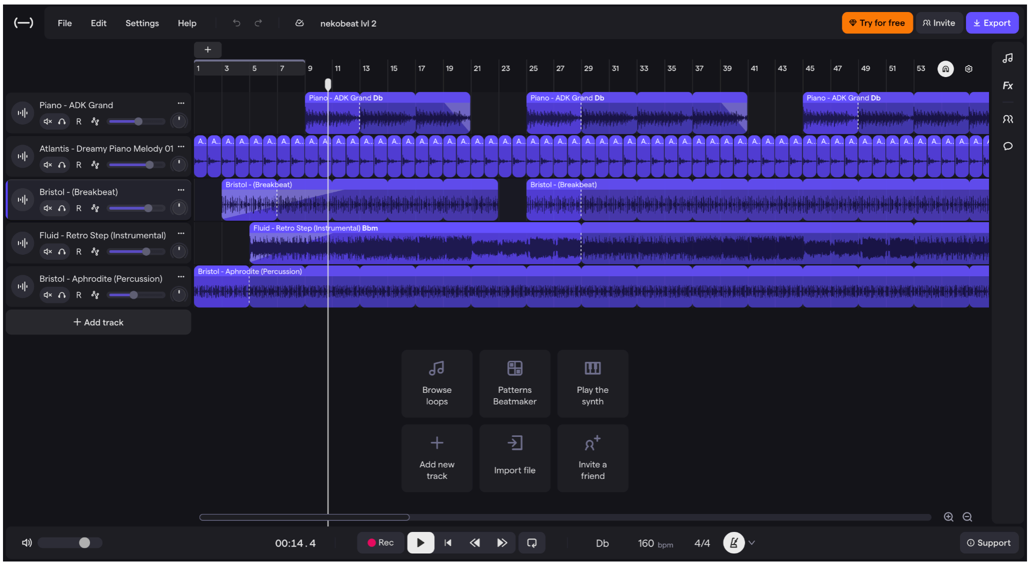
- "oh do i press before it hits the line, or after?"
- Expressed some difficulty getting used to timing
- Not enough allowance in hit bar
- Hard to hit exactly on bar
- Hard to get used to hit bar
- Had to just spam keys to hopefully hit on bar
- Rlly likes graphics/bg design
- Reminds of piano tiles
- Nothing confusing, pretty straightforward
- Not enough time to realize mistake before restarts
- Motivates rather than frustrates
- Integrate animations
- Adding score system/progress bar would be good

- Tap feedback on screen
- Include combos
- score based, reduced score for missed keys
- Power up items, obstacles
- Implement tremors in lvl 1, have to get on line but jumps up + down so its hard to get (basically hit bar is the tremors, hit bar jitters
- Add mechanics each lvl
- Indicator that remors shake (screen shakes a little maybe)
- Only karen has to get it, not general public
- Each lvl sho

Record key observations in a more informal way (this is not a UX study)

Version history
All versions
March 5, 3:56 PM
Ellis Byun
Nat Nguyen
March 5, 3:25 PM
Ellis Byun
Nat Nguyen
jasmine cabrera
March 5, 1:53 PM
Ellis Byun
March 5, 1:17 PM
Ellis Byun
March 4, 11:18 PM
Ellis Byun
March 4, 12:42 PM
Joanne Lang
March 2, 1:52 PM
jasmine cabrera
March 2, 12:05 PM
estelle
This month
March 1, 3:50 PM
Highlight changes





Entry #4: Lead Visual & Interaction Designer

Name: Nat Nguyen

Role(s): Lead Visual & Interaction Designer, Assistant Programmer

Primary responsibility for this work: Responsible for designing the visual elements of the game, including pixel characters, backgrounds, buttons, and logo.

Goal of Work Session

- Designed and created the game’s pixel characters (bunny and kitty)
- Drew and refined the game’s backgrounds to match the cyberpunk aesthetic, providing a visually engaging environment that supported the rhythm-based mechanics.
- Developed the game’s logo and interface buttons, focusing on clarity, visual hierarchy, and accessibility for players.

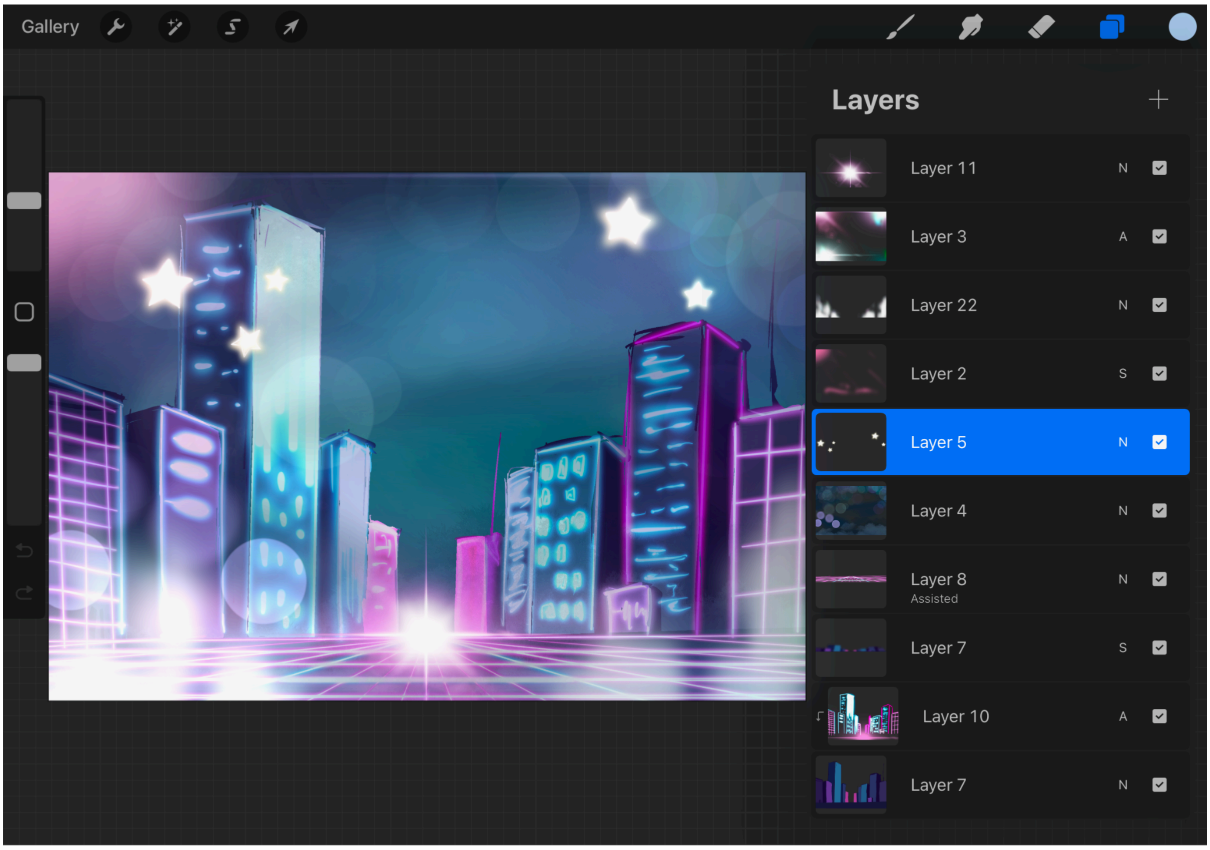
- Collaborated with the team to ensure visual elements aligned with research insights on disability framing

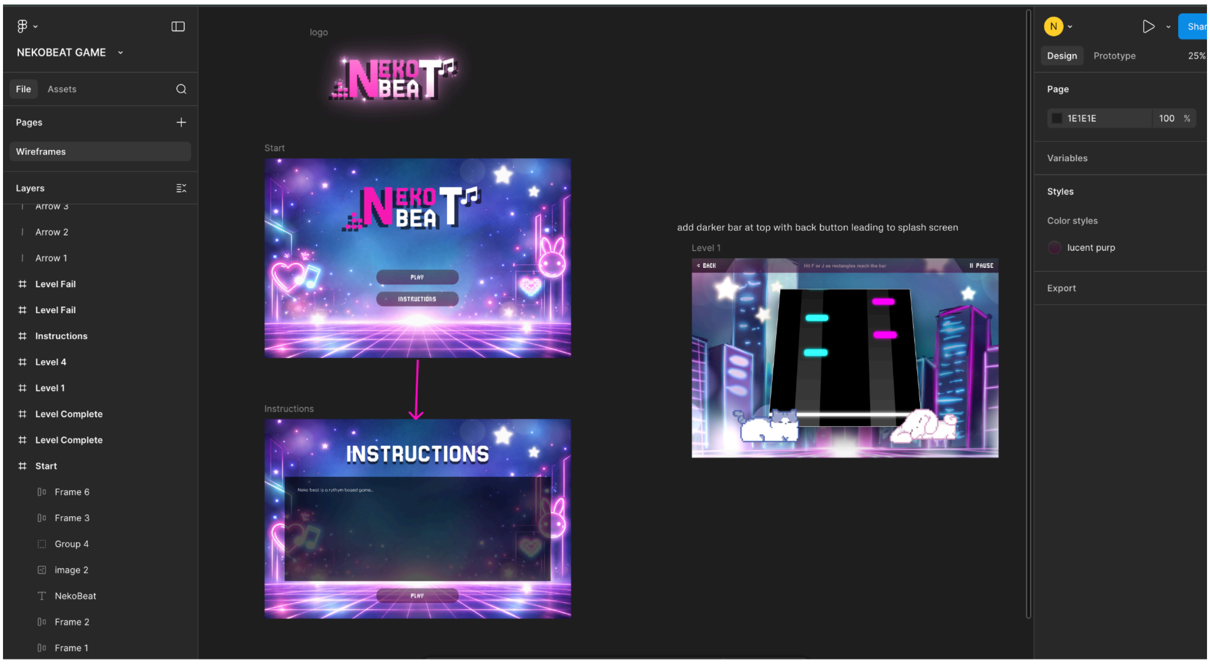
Tools, Resources, or Inputs Used

- Piskel for kitty and bunny characters
- Procreate for background creation
- Figma for buttons and logo

GenAI Documentation

No AI was used.





Entry #5: Initial Rhythm Game Ideation

Name: Joanne Lang

Role(s): Lead Systems Designer, Assistant Programmer

Primary responsibility for this work: Exploring feasibility of coding a rhythm game for our project.

Goal of Work Session

- Prompt GenAI to provide programmer team member (Jasmine) with a starting point or ideas of how to implement a rhythm game.

Tools, Resources, or Inputs Used

- ChatGPT 5.2

GenAI Documentation

Date Used: Mar. 2nd, 2026

Tool Disclosure: ChatGPT 5.2

Purpose of Use: Initial ideation and idea feasibility testing.

Summary of Interaction: Asked ChatGPT for a simple rhythm game code example in javascript, and it delivered a minimal code example.

Human Decision Point(s): Looked over the output to make sure there was nothing extremely wrong with it, then handed it off to Jasmine to take a look at and start iterating on.

Integrity & Verification Note: Looked over it and compared it to my knowledge and expertise of javascript coding and the course content.

Scope of GenAI Use: This interaction was very brief (just one prompt from me) and I just handed it off to Jasmine as a starting point for her and confirmation of the feasibility of a rhythm game. No need to iterate on it.

Limitations or Misfires: N/A, aside from the limited detail/complexity of the output code, but as a jumping off point it sufficed.

Summary of Process (Human + Tool)

- Looked over output and compared to existing coding knowledge
- Discussed validity with group members

Decision Points & Trade-offs

- Decided it was good enough as a jumping off point to give to Jasmine

Verification & Judgement

- Group discussion
- Comparison with course concepts
- Re-reading assignment criteria

Limitations, Dead Ends, or Open Questions

- Accepted the simplicity of the code because it was just to test what the code for a rhythm game could potentially look like
- Was left wondering what the code would look like actually implemented

Entry #6: Post-Playtesting Work Session

Name: Joanne Lang

Role(s): Lead Systems Designer, Assistant Programmer

Primary responsibility for this work: Implementing feedback from playtesting session, finishing up game functions and UI.

Goal of Work Session

- Implemented three refinements from playtest (increased allowance on hit bar, increased thickness of hit bar, introduced new game mechanic of moving bottom bar, longer duration of level)
- Further developed overall game by adding home and introduction screens, pause function, sound effects, and updating UI (fonts, buttons)

Tools, Resources, or Inputs Used

- ChatGPT 5.2 and ChatGPT 4.0
- Epidemic Sound for audio

GenAI Documentation

Everyone must complete this section. If not GenAI was used, write, “No GenAI used for this task.” When GenAI is not used, process evidence should still demonstrate iteration, revision, or development over time.

Because GenAI can closely mimic human-created work, instructors or TAs may occasionally request additional process evidence to confirm non-use. This may include original working files (e.g., an illustrator file), intermediate drafts, or a brief check-in with a TA to walk through your process.

These requests are not an assumption of misconduct. They are part of ensuring academic integrity in an environment where distinguishing between human-created and AI-generated work is increasingly difficult.

If GenAI was used (keep each response as brief as possible):

Date Used: Mar. 11, 2026

Tool Disclosure: ChatGPT 5.2 and ChatGPT 4.0

Purpose of Use:

- Requested wording improvements and explanations (e.g., clearer instructions or phrasing, comments explaining code snippets)
- Coding support and debugging
- Asked for help with implementing specific features, i.e. randomized falling rectangles, level timing logic, background switching, and sound effects
- UI/CSS adjustments, i.e. fonts, shadows, colors, and background image behavior
- Used it to troubleshoot technical issues, i.e. background music not playing due to browser autoplay restrictions

Summary of Interaction:

- Asked it to explain code written by Jasmine, inputting comments into the files to enhance my comprehension of the code she wrote
- Tool helped generate and modify js game logic, including timing for Level 1, randomized note spawning, lane restrictions for F and J keys, and hit detection behavior
- Suggested HTML and CSS updates to add fonts from Google Fonts, apply drop shadows, and configure different background images for different screens
- Provided audio integration guidance, including looping background music and adding sound effects for keyboard inputs
- Asked it for search terms and sources for sound effects to match a specific audio style (i.e. a short bass “bum” sound).

Human Decision Point(s):

- My initial approach was to provide all of the starting code as context, and tell it to implement something complex reflected across multiple files and output the updated code for all affected files — I realized after working like this that it was highly inefficient, and the results would fix one problem but cause another
- I switched to working on small code snippets and single features at a time, resulting in faster output generation time and more effective results

- I redirected the implementation of the level timing system, specifying that Level 1 should last 60 seconds with random rectangle spawns, a moving hit bar after 30 seconds, and a pause before Level 2, rather than ChatGPT's earlier approach
- I had to correct the lane logic, specifying that teal F notes must only appear in the left lane and purple J notes only in the right lane.
- I clarified that the background images should differ by screen, with bkgdGame.png for menu screens and cyberpunk.png only for the gameplay screen
- When the suggested audio implementation broke the key input mechanics, I redirected the solution so that sound effects would play without changing the existing key input behavior
- I refined requests to match my intended design choices, such as specifying fonts, color codes, and visual effects, and often disregarded ChatGPT's suggestions, instead making my own design decisions

Integrity & Verification Note:

- Comparison with course concepts
- Re-reading assignment criteria
- I would skim through the code output to see if there was anything glaringly off to me
- I would test it in VS code and Live Server to see if all of the functionalities worked and the UI looked good to make sure it didn't break a different part of the game (which it often did)

Scope of GenAI Use:

- Our group designed the overall concept of the game, i.e. game theme, mechanics, and structure, and I made the executive decision on design choices like colours, fonts, etc.
- I wrote and maintained the project structure, asset organization, and repository setup myself
- Our group selected and created the assets (images, audio files, fonts) used in the game
- I decided which suggestions to implement and manually integrated and tested code changes within the project
- The final game behavior and design resulted from iterative adjustments and human decisions, not direct copying of all generated outputs

Limitations or Misfires:

- Some responses did not initially solve the issue, requiring follow-up prompts i.e. background music not playing due to browser autoplay policies
- Certain code suggestions interfered with existing gameplay logic e.g. when adding sound effects prevented key inputs from hitting notes
- Some UI or CSS suggestions did not fully account for how the existing layout behaved e.g. background images appearing cut off
- Several responses required clarification or multiple iterations before producing a working solution
- The tool sometimes provided generalized solutions rather than ones tailored to my exact project structure, requiring additional adjustments on my part
- Didn't provide good search words for the sound effect I was looking for

Summary of Process (Human + Tool)

- I iteratively developed the game by testing features in the browser, identifying issues, and refining the implementation with the help of ChatGPT suggestions
- I used the tool to generate possible code changes for gameplay logic, UI styling, and audio integration, then tested those changes in my local project
- When something didn't work (e.g. background music not playing, sound effects interfering with key input), I returned to the tool with a more specific prompt, clarified the issue, and tried a revised approach
- I also experimented with visual design adjustments (fonts, color codes, shadows, and backgrounds) to align the game with the intended cyberpunk aesthetic
- This process involved multiple cycles of trial, testing, and revision, including moments where the initial solution failed and required reworking
- I consulted with my group members throughout the process, sending them videos of my implementations to get their feedback and iterating using it
- Biggest moment of uncertainty/failure would likely be trying to integrate background music and debugging errors during that process; had to try many different approaches and refine my ChatGPT prompts to troubleshoot successfully

Decision Points & Trade-offs

Describe one or two key decisions you made:

- What options you considered
- What changed
- Why that choice was made

These decisions should align with decision points shown in your visualization (for A1 – A3).

1. Structuring Level 1 as a time-based challenge

   Options considered:

   - Ending the level after a fixed number of falling rectangles
   - Using a time-based structure with randomized spawns

   Decision: I implemented a 60-second Level 1 with randomized note spawning, with an additional difficulty change (moving hit bar) after 30 seconds.

   Reason: A time-based approach made the gameplay feel more like a rhythm game and less predictable, improving pacing and player engagement.

2. Separating note types by lane

   Options considered:

   - Allowing both teal and purple rectangles to spawn in either lane
   - Assigning F notes to the left lane and J notes to the right lane

   Decision: I restricted each note type to a specific lane.

   Reason: This simplified the controls and improved clarity for the player, making it easier to learn the rhythm and avoid confusion during gameplay.

Verification & Judgement

Explain how you evaluated whether your decision or change was appropriate:

Examples:

- Playtesting
- Group discussion
- Peer or instructor feedback
- Comparison with course concepts
- Re-reading assignment criteria
- Ethical, representational, accessible, or global considerations

- I verified most decisions through manual playtesting, checking whether features behaved as intended (e.g. timing of falling rectangles, hit detection, sound playback)
- When changes were implemented, I ran the game and tested key interactions to confirm that inputs, visuals, and audio all worked together

- I evaluated whether design changes improved clarity and usability, such as lanes for different keys or adjusting visual feedback for hits
- I referred to assignment expectations for a functional interactive game, making sure the features contributed to a playable experience that successfully integrated our assigned disability

Limitations, Dead Ends, or Open Questions

Note anything that:

- Did not work as expected
- Was left unresolved

Required compromise or simplification

- Some features did not work on the first attempt i.e. background music playback, which required additional troubleshooting due to browser autoplay restrictions
- Adding sound effects initially interfered with the key input logic, requiring iteration so the game mechanics continued functioning correctly
- Some UI adjustments, e.g. background images, did not behave as expected at first and needed further CSS adjustments
- Some aspects of audio design (e.g. finding the right subtle bass “bum” sound) were still exploratory, and the final sound choice required additional refinement
- Due to time constraints, some features were implemented in a simplified form, prioritizing a working gameplay loop over more complex audio or visual polish

Entry #7: Post-Midterm Game Showcase Work Session

Name: Joanne Lang

Role(s): Lead Systems Designer, Assistant Programmer

Primary responsibility for this work: Update and finalize game for A2 submission, implementing feedback from Midterm Showcase.

Goal of Work Session

- Sound effects: the synth/laser sound effect when pressing keys became jarring and distracting after repetitive hits during gameplay — FINALLY found a better, subtler sound effect (referred to as “error beep” in Epidemic Sound)
- Visual alignment: the blue cat character appeared too close to the falling bars, affecting the visual balance — I shifted it further to the left

Tools, Resources, or Inputs Used

- Epidemic Sound for audio

GenAI Documentation

No GenAI used for this task.

Appendix

Input (Estelle):

Attached Assignment 2 Outline PDF and Group Charter

My primary role for this assignment is Lead Experience Designer. I am about to begin research on both Parkinson's disease and disability representation in video games. As a starting point, provide some important topics that I need for the basis of my research.

Output (ChatGPT 5.2):

Perfect — now you're stepping fully into your **Lead Experience Designer** role.

Since your game must teach non-disabled players about **Parkinson's disease**, your research should not just be medical. It needs to support:

- Empathy
- Mechanic translation
- Ethical framing
- Playtesting evaluation

Here's exactly what you should research — organized by priority.

🔴 1. Core Understanding of Parkinson's (Foundation)

You need a strong baseline understanding before designing anything.

Research:

- What Parkinson's disease is (neurological basis)
- Dopamine's role in motor control
- How symptoms progress over time
- Typical age of onset
- Variability between patients

You are NOT becoming a doctor — you are learning enough to avoid misrepresentation.

🟠 2. Motor Symptoms (Mechanic Translation Goldmine)

These are most directly translatable into gameplay mechanics.

Research in detail:

- Tremors (resting tremor vs action tremor)
- Bradykinesia (slowness of movement)
- Rigidity (muscle stiffness)
- Postural instability (balance issues)
- Freezing episodes

For each symptom, ask:

- What does it feel like?
- Is it predictable or unpredictable?
- Is it constant or intermittent?
- Is it frustrating? Embarrassing? Fatiguing?

This directly informs mechanics like:

- Input delay
- Screen shake
- Movement resistance
- Random interruptions

🟡 3. Non-Motor Symptoms (Often Overlooked but Powerful)

This is where empathy deepens beyond “shaky hands.”

Research:

- Cognitive slowing
- Speech changes
- Fatigue
- Depression
- Anxiety
- Loss of facial expression (masked face)

These can inspire:

- Communication barriers
- Time pressure tension

- Misinterpretation mechanics
- Emotional misalignment

If you only focus on tremors, your game may feel shallow.

🟢 4. Lived Experiences (Most Important for Empathy)

This is critical for your role.

Look for:

- First-person accounts
- Interviews
- Patient blogs
- TED talks
- Documentary clips

Search questions like:

- “What does Parkinson’s feel like?”
- “Daily life with Parkinson’s”
- “Most frustrating part of Parkinson’s”

Pay attention to:

- Emotional language
- Social challenges
- Identity shifts
- Loss of independence

This informs how your game should feel — not just function.

🔵 5. Misconceptions & Stereotypes

Very important for ethical design.

Research:

- Common public misunderstandings
- Media portrayals of Parkinson’s

- Harmful tropes

You want to avoid:

- Making it comedic
- Reducing it to shaking
- Treating it as only physical

This protects your design from being insensitive.

🟣 6. Existing Games About Illness/Empathy

Research similar games that teach through mechanics.

Examples to look at:

- **That Dragon, Cancer** (illness empathy)
- **Before Your Eyes** (embodied mechanic design)
- **SPENT** (economic hardship simulation)

Research:

- How mechanics reflect lived experience
- How frustration is used intentionally
- How they avoid being preachy

This strengthens your conceptual grounding.

🟤 7. Ethical Representation in Game Design

Look into:

- Disability representation in media
- Simulation vs exploitation
- Empathy vs pity
- The concept of “serious games”

Ask:

- How do we design challenge without trivializing suffering?

- How do we avoid making disability a gimmick?

This directly shapes your responsibility as Experience Designer.

⚫ 8. Teaching Outcome Clarity

Since A2 requires a demonstrable teaching outcome, research:

- What do non-disabled people typically not understand about Parkinson’s?
- What insight would actually change perspective?

Example outcomes:

- “Motor control is not always voluntary.”
- “Simple tasks require immense effort.”
- “Symptoms fluctuate unpredictably.”

Your game must clearly communicate ONE of these.

🎯 How to Organize Your Research Notes

Create sections:

1. Medical overview
2. Motor symptoms (mechanic ideas beside each)
3. Non-motor symptoms (mechanic ideas beside each)
4. Lived experience quotes
5. Misconceptions to avoid
6. Potential learning outcomes

This will make mechanic brainstorming much easier later.

🚨 Important Reminder

Your job is not:
“Make the game hard.”

Your job is:
"Make the player understand something meaningful."

That difference is everything.

If you'd like, I can next:

- Help you turn research into mechanic ideas
- Help you narrow which symptoms are most feasible for p5.js
- Or help you craft a strong teaching outcome statement.

Joanne Entry #5: Initial Rhythm Game Ideation
Transcript:
https://chatgpt.com/share/69b3894f-f9e8-8003-8680-291f9a2ef711

Joanne Entry #6: Post-Playtesting Work Session
Transcript:
https://chatgpt.com/share/69b378e0-6484-8003-84ea-7e4221b1a222

References

[9] Charity, AbleGamers. “Video Games and Disability Representation.” The AbleGamers Charity, 2023. https://ablegamers.org/video-games-disability-representation/.

[10] Epidemic Sound. n.d. "User Interface, Beep, Error Tone, Soft." Available at: https://www.epidemicsound.com/sound-effects/tracks/17f72d25-b4a6-4d8f-addc-f446ee8b3cb7/.

[6] Fox, Michael J. “The Michael J. Fox Foundation for Parkinson’s Research.” The Michael J. Fox Foundation for Parkinson’s Research | Parkinson’s Disease, 2019. https://www.michaeljfox.org/.

[7] González, Julia, Leya George, Lidia Miteva, and Aneesha Singh. “Developing Empathy towards Experiences of Invisible Disabilities through Games.” In *EmpathiCH ’23: Proceedings of the 2nd Empathy-Centric Design Workshop*, 1–8. University College London, 2023. https://doi.org/10.1145/3588967.3588976.

[8] Karen Anne Cochrane and David Han. 2026. Week 4 - Part 1. GBDA302 Course Slides. University of Waterloo, Stratford, Canada.

[2] Mayo Clinic. “Parkinson’s Disease.” Mayo Clinic, September 27, 2024. https://www.mayoclinic.org/diseases-conditions/parkinsons-disease/symptoms-causes/syc-20376055.

[1] National Institute of Neurological Disorders and Stroke. “Parkinson’s Disease: Challenges, Progress, and Promise.” www.ninds.nih.gov, 2023. https://www.ninds.nih.gov/current-research/focus-disorders/parkinsons-disease-research/parkinsons-disease-challenges-progress-and-promise.

[3] Parkinson's Foundation. “Movement Symptoms.” Parkinson’s Foundation, 2017. https://www.parkinson.org/Understanding-Parkinsons/Movement-Symptoms.

[4] Parkinson's Foundation. “Non-Movement Symptoms.” Parkinson’s Foundation, 2023. https://www.parkinson.org/Understanding-Parkinsons/Non-Movement-Symptoms.

[5] TEDx Talks. “My Parkinson’s Story: Lessons on Optimism and Opportunity | Mark Colo.” *Youtube*, 2023. https://youtu.be/asf-YXe7hXk?si=2Vrr3nJTfbasGO85.